SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

Beginner

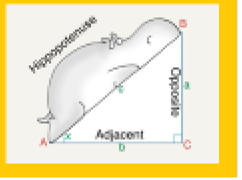
TRIGONOMETRIC FUNCTIONS

RIGHT TRIANGLE APPROACH

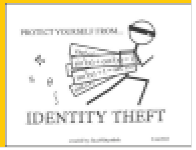
Session outline

Trigonometry: introduction

Trigonometry of Right Triangles



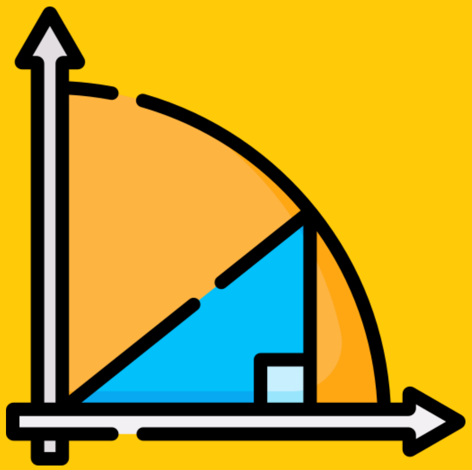
Trigonometric Identities



Learning Objectives

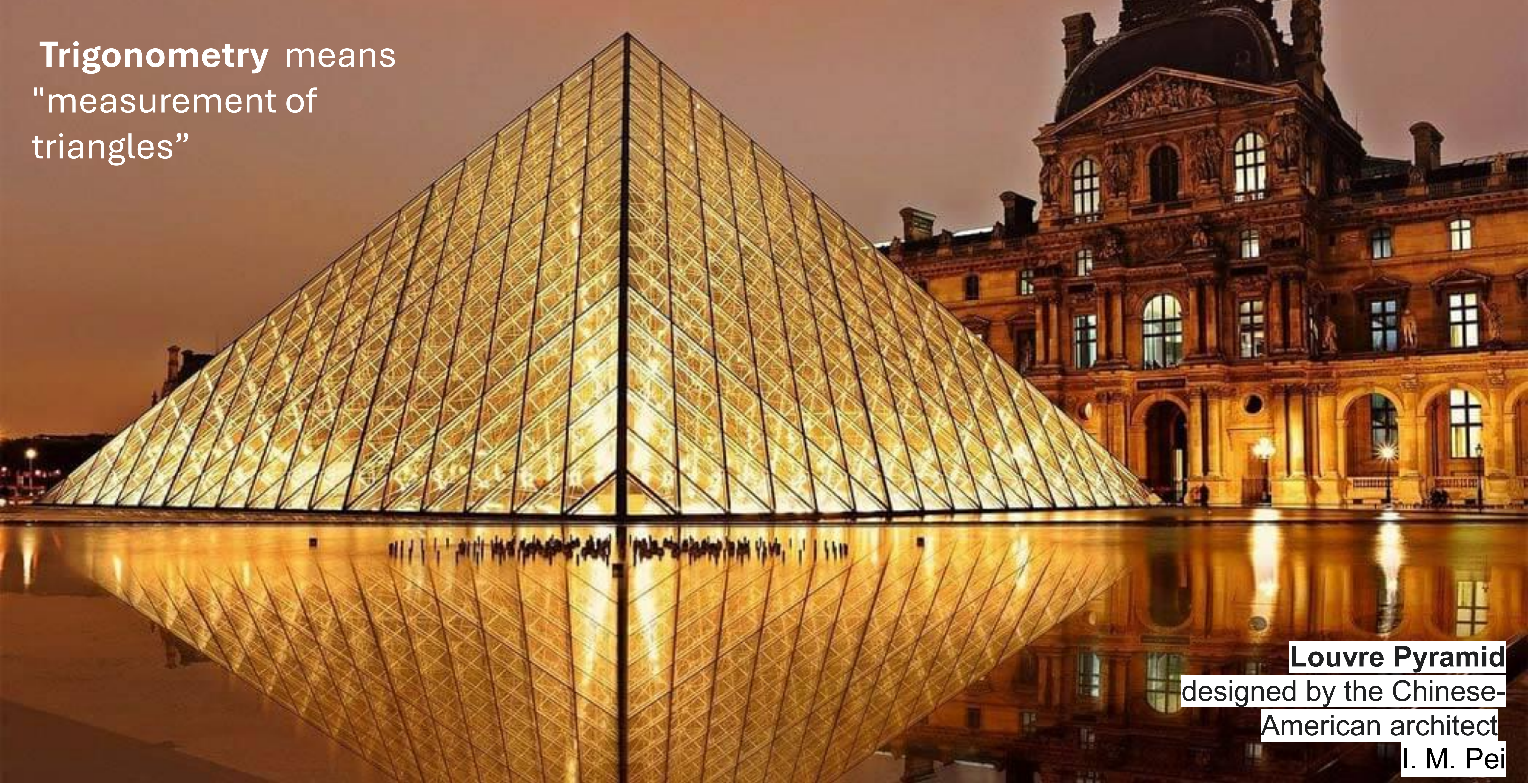
- Revise the trigonometry of right triangles, including the **Pythagorean Theorem** and trigonometric ratios.
- Understand **trigonometric functions for special triangles** and memorize special values of trigonometric functions.
- Solve word problems involving trigonometric calculations, such as finding heights and distances.
- Understand and apply trigonometric identities to simplify expressions and solve problems.
- **Express one trigonometric function in terms of another** and evaluate trigonometric functions in different quadrants.

Trigonometry: introduction



What is Trigonometry and where is it used?

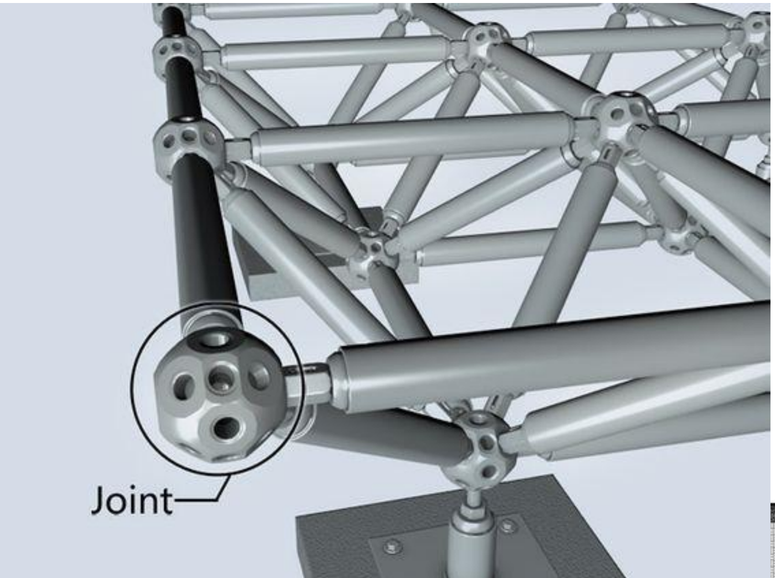
Trigonometry means "measurement of triangles”

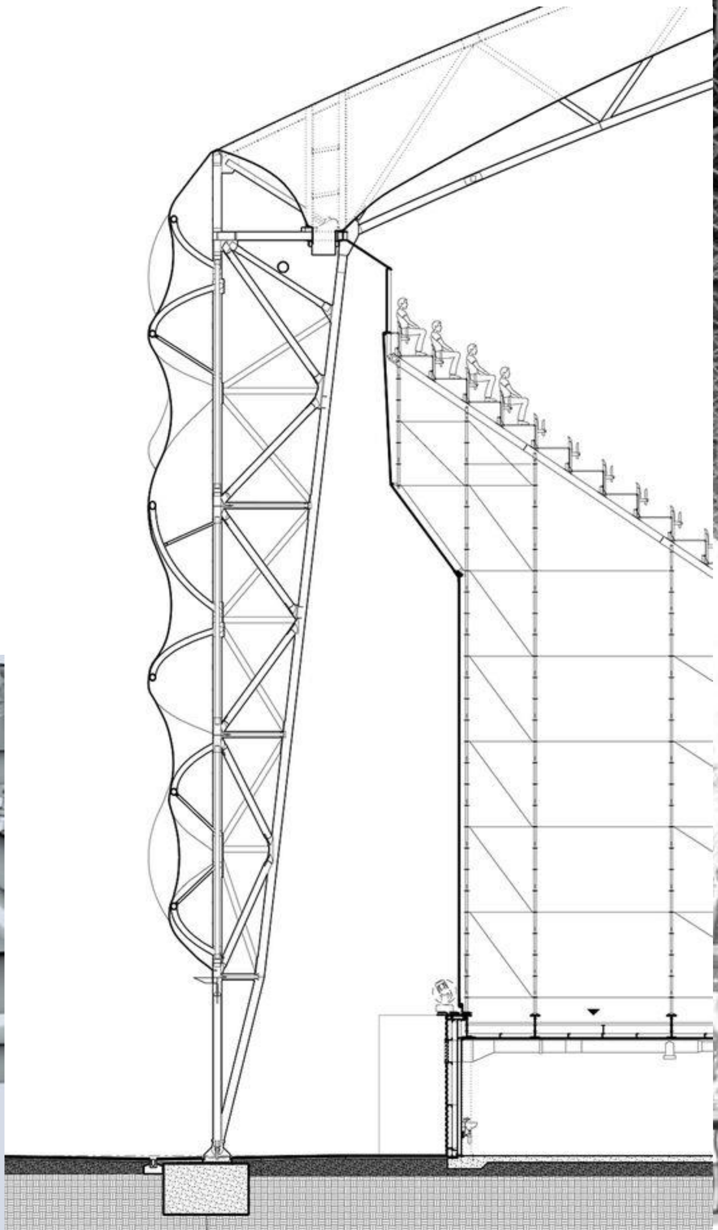


Louvre Pyramid
designed by the Chinese-American architect
I. M. Pei

Trigonometry in Engineering

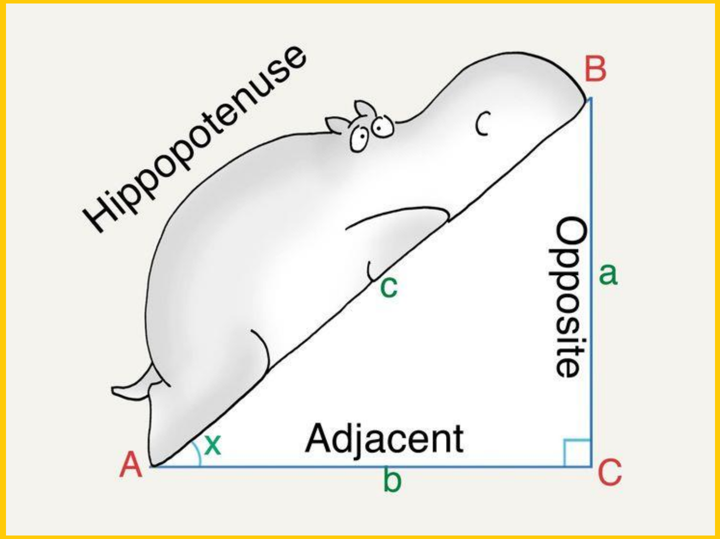
How would you calculate roof slopes, ground surfaces, light angles, structural loads, and height and width of structures to design buildings?



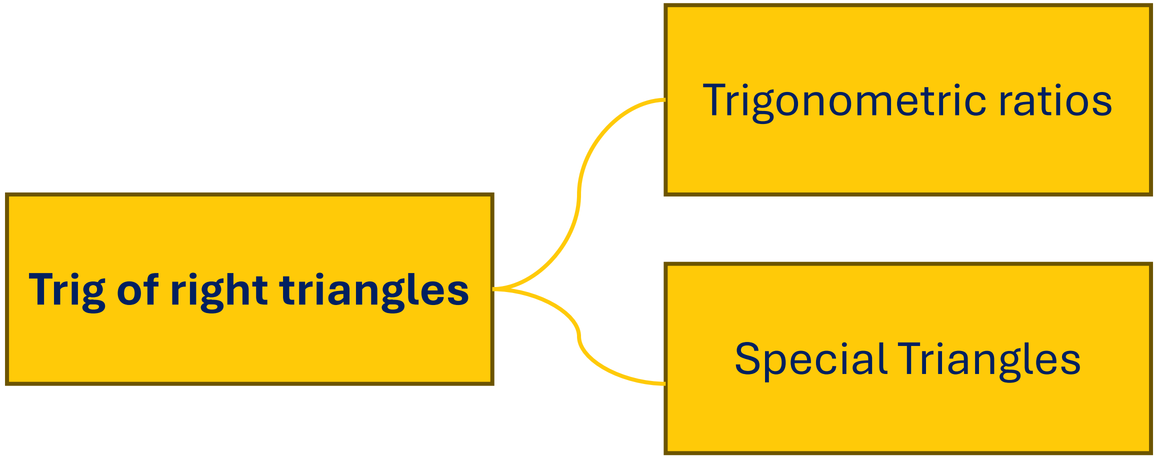


Ferrari World Yas Island, Abu Dhabi

Trigonometry of Right Triangles



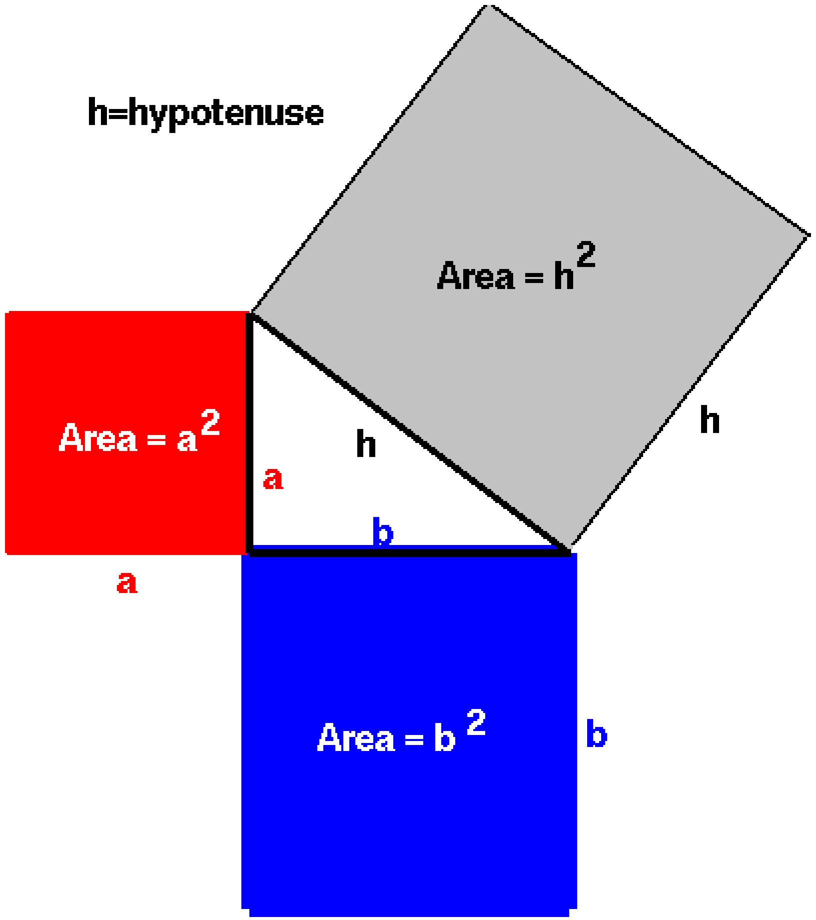
Trig of right triangles: Concept map



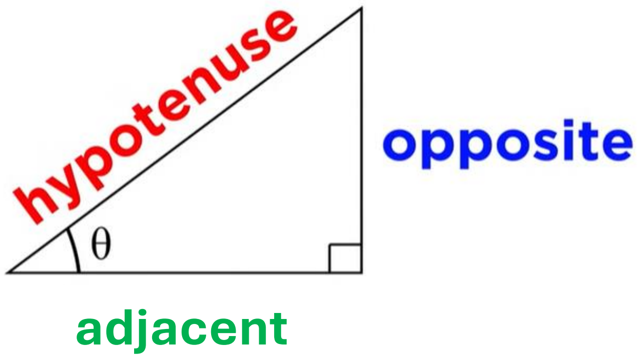
Pythagorean Theorem

For any right triangle with sides a and b and hypotenuse h, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$h^2 = a^2 + b^2$$



Trigonometric ratios



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

"sine of theta"

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

"cosine of theta"

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

"tangent of theta"

mnemonic

SOH

CAH

TOA

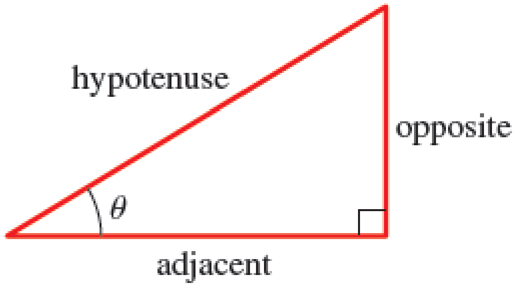
Six Trigonometric Ratios

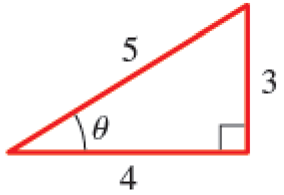
THE TRIGONOMETRIC RATIOS

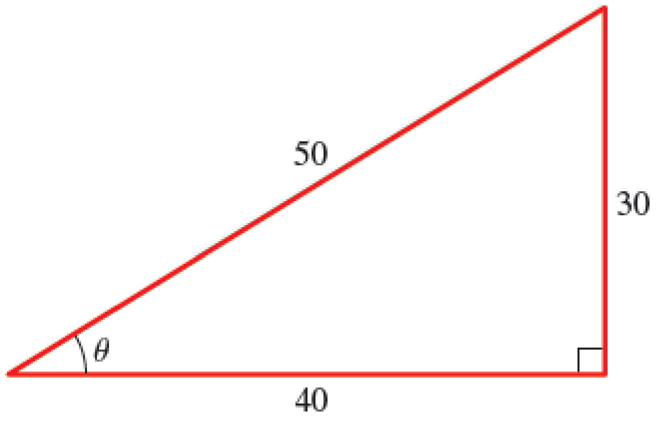
$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}} \qquad \sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}} \qquad \cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$

"cosecant" "secant" "cotangent"







$\sin\theta = \frac{3}{5}$ $\qquad$ $\sin\theta = \frac{30}{50} = \frac{3}{5}$

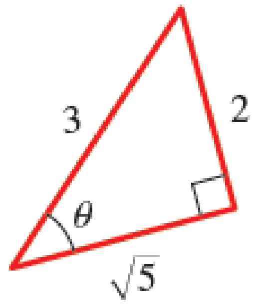
Same!

Since any two right triangles with same angle θ are similar,
→these ratios are the **same**, **regardless of the size** of the triangle

trigonometric ratios **depend only on the angle θ**

Finding Trigonometric Ratios

Find the six trigonometric ratios of the angle θ in the figure



SOLUTION By the definition of trigonometric ratios, we get

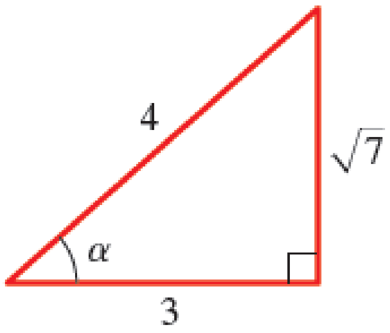
$$\sin\theta = \frac{2}{3} \qquad \cos\theta = \frac{\sqrt{5}}{3} \qquad \tan\theta = \frac{2}{\sqrt{5}}$$

$$\csc\theta = \frac{3}{2} \qquad \sec\theta = \frac{3}{\sqrt{5}} \qquad \cot\theta = \frac{\sqrt{5}}{2}$$

EXAMPLE

Finding Trigonometric Ratios

If $\cos\alpha = \frac{3}{4}$, sketch a right triangle with acute angle α, and find the other five trigonometric ratios of α.



SOLUTION Since $\cos\alpha$ is defined as the ratio of the adjacent side to the hypotenuse, we sketch a triangle with hypotenuse of length 4 and a side of length 3 adjacent to α. If the opposite side is x, then by the Pythagorean Theorem, $3^2 + x^2 = 4^2$ or $x^2 = 7$, so $x = \sqrt{7}$. We then use the triangle in Figure 4 to find the ratios.

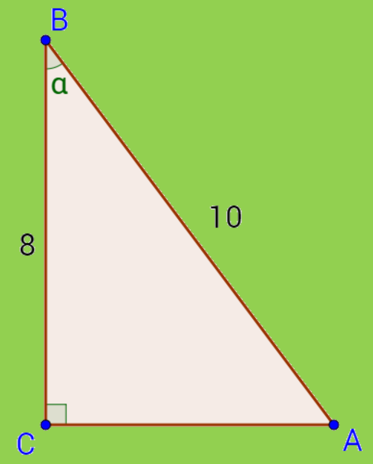
$$\sin\alpha = \frac{\sqrt{7}}{4} \qquad \cos\alpha = \frac{3}{4} \qquad \tan\alpha = \frac{\sqrt{7}}{3}$$

$$\csc\alpha = \frac{4}{\sqrt{7}} \qquad \sec\alpha = \frac{4}{3} \qquad \cot\alpha = \frac{3}{\sqrt{7}}$$

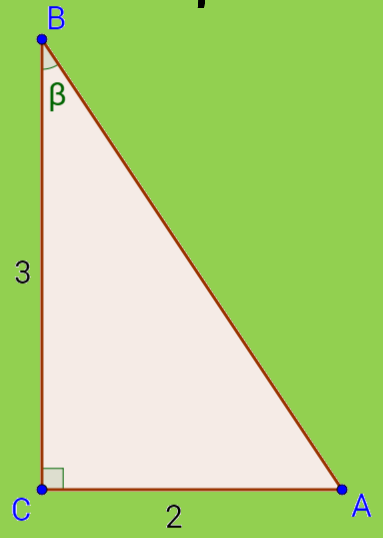
EXAMPLE

1. Finding trigonometric ratios

a. Evaluate the six trigonometric functions of α for the triangle in the figure



b. Evaluate the six trigonometric functions of β for the triangle in the figure



ACTIVITY 1

4a. Finding trigonometric ratios

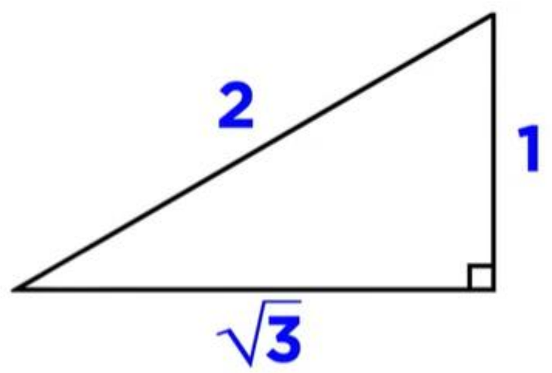
SOLUTION

4b. Finding trigonometric ratios

SOLUTION

Evaluating Trig functions for special triangles

30-60-90 triangle



30°

$\sin\theta = 1/2$

$\cos\theta = \sqrt{3}/2$

$\tan\theta = \sqrt{3}/3$

By removing the root in the denominator

$1/\sqrt{3} \cdot (\sqrt{3}/\sqrt{3})$

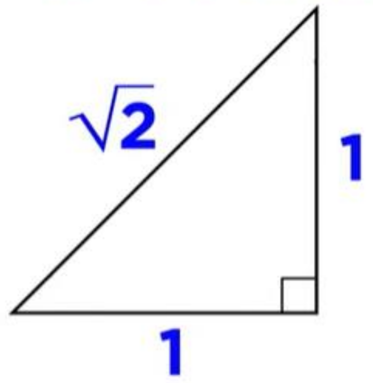
60°

$\sin\theta = \sqrt{3}/2$

$\cos\theta = 1/2$

$\tan\theta = \sqrt{3}$

45-45-90 triangle



45°

$\sin\theta = \sqrt{2}/2$

$\cos\theta = \sqrt{2}/2$

By removing the root in the denominator

$1/\sqrt{2} \cdot (\sqrt{2}/\sqrt{2})$

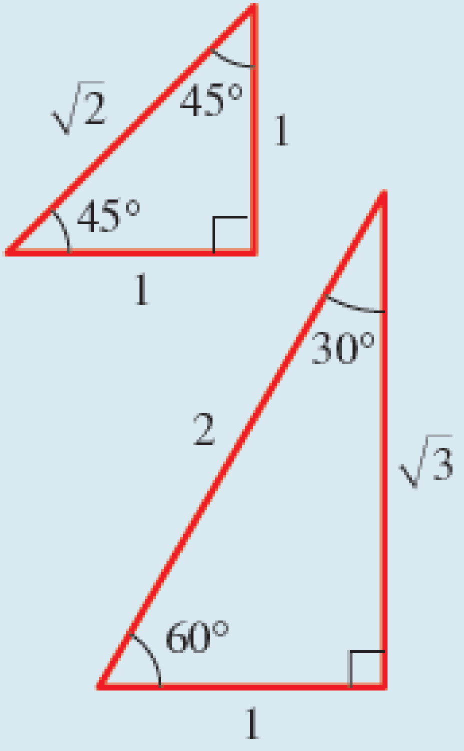
$\tan\theta = 1$

Special values of Trigonometric functions

SPECIAL VALUES OF THE TRIGONOMETRIC FUNCTIONS

The following values of the trigonometric functions are obtained from the special triangles.

θ in degrees	θ in radians	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
0	0	0	1	0	—	1	—
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
90°	$\frac{\pi}{2}$	1	0	—	1	—	0



Good to memorize these!

Solving a Right Triangle

Solve triangle *ABC*, shown in Figure

SOLUTION It's clear that $\angle B = 60°$

$$\sin 30° = \frac{a}{12} \qquad \text{Definition of sine}$$

$$a = 12 \sin 30° \qquad \text{Multiply by 12}$$

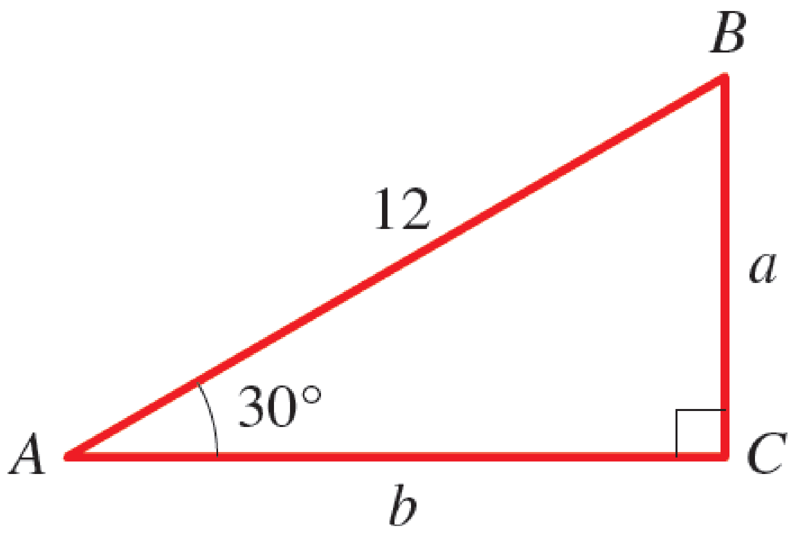
$$= 12\left(\tfrac{1}{2}\right) = 6 \qquad \text{Evaluate}$$

Also from Figure 7 we have

$$\cos 30° = \frac{b}{12} \qquad \text{Definition of cosine}$$

$$b = 12 \cos 30° \qquad \text{Multiply by 12}$$

$$= 12\left(\frac{\sqrt{3}}{2}\right) = 6\sqrt{3} \qquad \text{Evaluate}$$



EXAMPLE

Finding heights

EXAMPLE 5 ■ Finding the Height of a Tree

A giant redwood tree casts a shadow 532 ft long. Find the height of the tree if the angle of elevation of the sun is 25.7°.

SOLUTION Let the height of the tree be h. From Figure 10 we see that

$$\frac{h}{532} = \tan 25.7° \qquad \text{Definition of tangent}$$

$$h = 532 \tan 25.7° \qquad \text{Multiply by 532}$$

$$\approx 532(0.48127) \approx 256 \qquad \text{Use a calculator}$$

Therefore the height of the tree is about 256 ft.

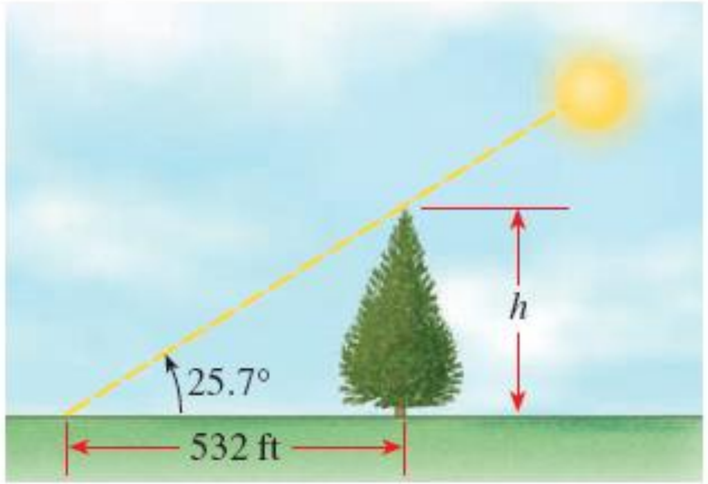


FIGURE 10

EXAMPLE

A plane is flying within sight of an airport, at an elevation of 10,500 m. The pilot would like to estimate the plane's distance to a point on the runway and finds that the angle of depression to that point is 22 degrees .

(a) What is the distance between the plane and the point on the runway?
(b) What is the distance between a point on the ground directly below the plane and the point on the runway?

WORD PROBLEM 1

Word problem 1

SOLUTION

Elevation of a Kite A tourist is lying on the beach, flying a kite. The tourist holds the end of the kite string at ground level and estimates the angle of elevation of the kite to be $50°$. If the string is 135 m long, how high is the kite above the ground?

WORD PROBLEM 2

Word problem 2

SOLUTION

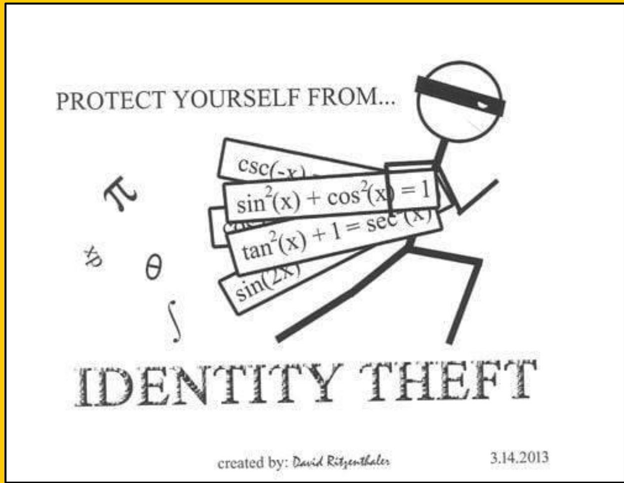
Height of a Balloon A hot-air balloon is floating above a straight road. To estimate their height above the ground, the balloonists simultaneously measure the angle of depression to two consecutive mileposts on the road on the same side of the balloon. The angles of depression are found to be 20° and 22°. How high is the balloon?

WORD PROBLEM 3

Word problem 3

SOLUTION

Trigonometric Identities



What is an identity?

An identity is an **equation** which is always **true**,
no matter what values are substituted

What is a trigonometric identity?

Trigonometric Identities are equalities
with **trigonometry functions**
that are
true
for all the values of variables
for which both sides of the equality are defined.

Identities help us in **simplifying expressions**.

Reciprocal identities

$$\sin\theta = \frac{opp}{hyp} \rightarrow \csc\theta = \frac{hyp}{opp} \qquad \sin\theta = \frac{1}{\csc\theta} \rightarrow \csc\theta = \frac{1}{\sin\theta}$$

$$\cos\theta = \frac{adj}{hyp} \rightarrow \sec\theta = \frac{hyp}{adj} \qquad \cos\theta = \frac{1}{\sec\theta} \rightarrow \sec\theta = \frac{1}{\cos\theta}$$

$$\tan\theta = \frac{opp}{adj} \rightarrow \cot\theta = \frac{adj}{opp} \qquad \tan\theta = \frac{1}{\cot\theta} \rightarrow \cot\theta = \frac{1}{\tan\theta}$$

Tangent and Cotangent

$$\boxed{\tan\theta = \frac{\sin\theta}{\cos\theta}}$$

$$\boxed{\cot\theta = \frac{\cos\theta}{\sin\theta}}$$

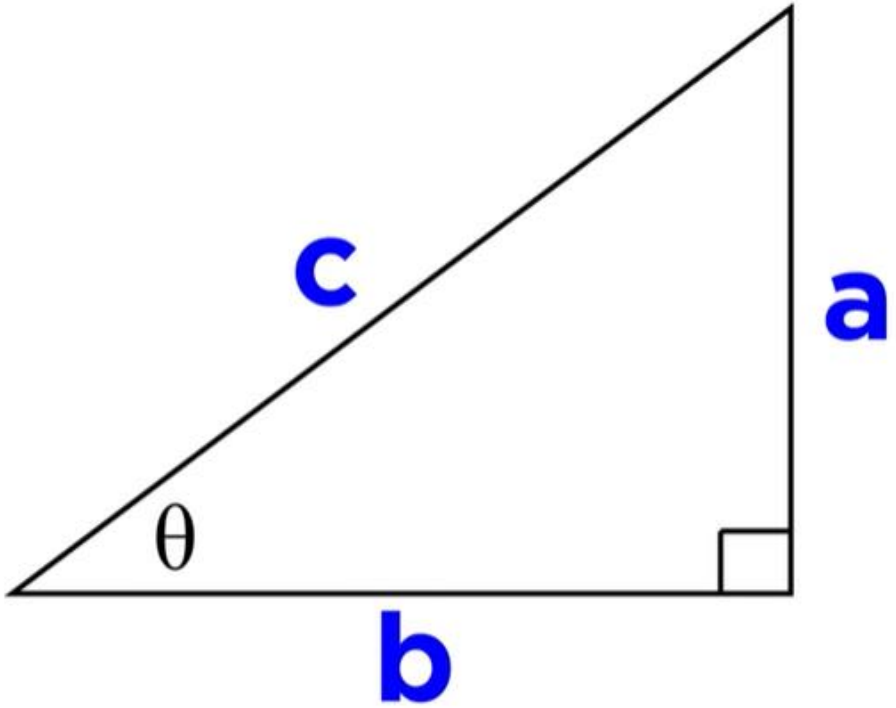
$$\tan\theta = \frac{opp/hyp}{adj/hyp}$$

$$\cot\theta = \frac{adj/hyp}{opp/hyp}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$

The Pythagorean identity-1



$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2}$$

$$(\sin\theta)^2 + (\cos\theta)^2 = 1$$

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

The Pythagorean Identity

CONVENTION: We write $sin^{2}t$ for $(\sin t)^{2}$

The Pythagorean identity-2

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

$$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

$$\tan^2\theta + 1 = \sec^2\theta$$

The Pythagorean identity-3

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

$$\frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$$

$$1 + \cot^2\theta = \csc^2\theta$$

Trigonometric Identities

FUNDAMENTAL IDENTITIES

Reciprocal Identities

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1 \qquad \tan^2\theta + 1 = \sec^2\theta \qquad 1 + \cot^2\theta = \csc^2\theta$$

Proof Let's prove the first Pythagorean identity. Using $x^2 + y^2 = r^2$ (the Pythagorean Theorem) in Figure 13, we have

$$\sin^2\theta + \cos^2\theta = \left(\frac{y}{r}\right)^2 + \left(\frac{x}{r}\right)^2 = \frac{x^2 + y^2}{r^2} = \frac{r^2}{r^2} = 1$$

Thus $\sin^2\theta + \cos^2\theta = 1$. (Although the figure indicates an acute angle, you should check that the proof holds for all angles θ.) ■

Expressing one Trigonometric Function in terms of another

(a) Express $\sin\theta$ in terms of $\cos\theta$.

SOLUTION

(a) From the first Pythagorean identity we get

$$\sin\theta = \pm\sqrt{1-\cos^2\theta}$$

where the sign depends on the quadrant. If θ is in Quadrant I or II, then $\sin\theta$ is positive, so

$$\sin\theta = \sqrt{1-\cos^2\theta}$$

whereas if θ is in Quadrant III or IV, $\sin\theta$ is negative, so

$$\sin\theta = -\sqrt{1-\cos^2\theta}$$

EXAMPLE

Expressing one Trigonometric Function in terms of another

(b) Express $\tan\theta$ in terms of $\sin\theta$, where θ is in Quadrant II.

SOLUTION

(b) Since $\tan\theta = \sin\theta/\cos\theta$, we need to write $\cos\theta$ in terms of $\sin\theta$. By part (a)

$$\cos\theta = \pm\sqrt{1 - \sin^2\theta}$$

and since $\cos\theta$ is negative in Quadrant II, the negative sign applies here. Thus

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sin\theta}{-\sqrt{1 - \sin^2\theta}}$$

EXAMPLE

Evaluating Trig functions

If $\tan\theta = \frac{2}{3}$ and θ is in Quadrant III, find $\cos\theta$.

SOLUTION 1 We need to write $\cos\theta$ in terms of $\tan\theta$. From the identity $\tan^2\theta + 1 = \sec^2\theta$ we get $\sec\theta = \pm\sqrt{\tan^2\theta + 1}$. In Quadrant III, $\sec\theta$ is negative, so

$$\sec\theta = -\sqrt{\tan^2\theta + 1}$$

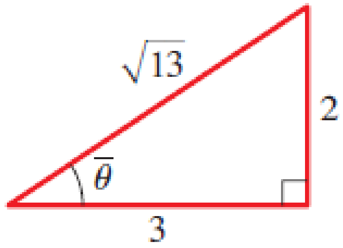
Thus

$$\cos\theta = \frac{1}{\sec\theta} = \frac{1}{-\sqrt{\tan^2\theta + 1}}$$

$$= \frac{1}{-\sqrt{\left(\frac{2}{3}\right)^2 + 1}} = \frac{1}{-\sqrt{\frac{13}{9}}} = -\frac{3}{\sqrt{13}}$$

If you wish to rationalize the denominator, you can express $\cos\theta$ as

$$-\frac{3}{\sqrt{13}} \cdot \frac{\sqrt{13}}{\sqrt{13}} = -\frac{3\sqrt{13}}{13}$$



SOLUTION 2 This problem can be solved more easily by using the method of Example 2 of Section 6.2. Recall that, except for sign, the values of the trigonometric functions of any angle are the same as those of an acute angle (the reference angle). So, ignoring the sign for the moment, let's sketch a right triangle with an acute angle $\bar{\theta}$ satisfying $\tan\bar{\theta} = \frac{2}{3}$ (see Figure 14). By the Pythagorean Theorem the hypotenuse of this triangle has length $\sqrt{13}$. From the triangle in Figure 14 we immediately see that $\cos\bar{\theta} = 3/\sqrt{13}$. Since θ is in Quadrant III, $\cos\theta$ is negative, so

$$\cos\theta = -\frac{3}{\sqrt{13}}$$

EXAMPLE

Evaluating Trig functions

If $\sec\theta = 2$ and θ is in Quadrant IV, find the other five trigonometric functions of θ.

SOLUTION We sketch a triangle as in Figure 15 so that $\sec\bar{\theta} = 2$. Taking into account the fact that θ is in Quadrant IV, we get

$$\sin\theta = -\frac{\sqrt{3}}{2} \qquad \cos\theta = \frac{1}{2} \qquad \tan\theta = -\sqrt{3}$$

$$\csc\theta = -\frac{2}{\sqrt{3}} \qquad \sec\theta = 2 \qquad \cot\theta = -\frac{1}{\sqrt{3}}$$

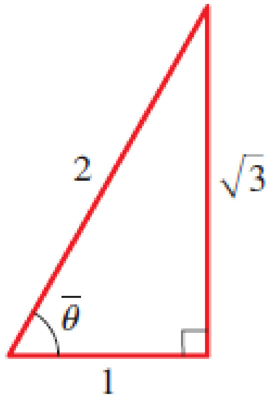


FIGURE 15

EXAMPLE

2. Expressing One Trigonometric Function in Terms of another

Write the first trigonometric function in terms of the second for θ in the given quadrant

a) $\tan\theta$, $\cos\theta$; θ in Quadrant III

b) $\sec\theta$, $\sin\theta$; θ in Quadrant I

ACTIVITY 2

Expressing One Trigonometric Function in Terms of Another

SOLUTION

Expressing One Trigonometric Function in Terms of Another

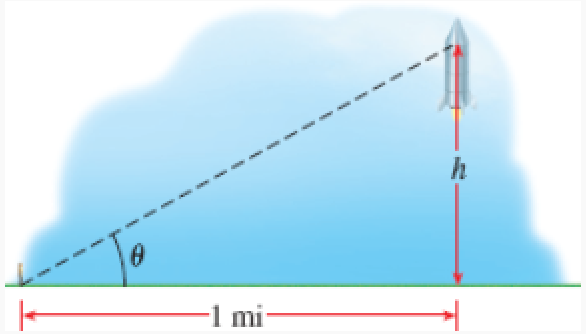
SOLUTION

Height of a Rocket A rocket fired straight upward is tracked by an observer on the ground, 1.6 km away.

(a) Show that when the angle of elevation is θ, the height of the rocket (in m) is $h = 1600 \tan \theta$.

(b) Complete the table to find the height of the rocket at the given angles of elevation.

θ	20°	60°	80°	85°
h				



WORD PROBLEM 4

Word problem 4

SOLUTION

Wooden Beam A rectangular beam is to be cut from a cylindrical log of diameter 20 cm. The figures show different ways this can be done.

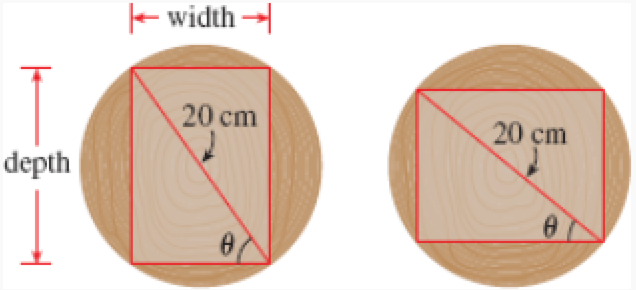
(a) Express the cross-sectional area of the beam as a function of the angle θ in the figures.

Answer

(b) Graph the function you found in part (a).

Answer

(c) Find the dimensions of the beam with largest cross-sectional area.



WORD PROBLEM 5

Word problem 5

SOLUTION

Summary

We covered

- Angle measure (Self study)
- Trigonometry of right triangles
- Trigonometric functions of angles (Self study)
- Trigonometric identities.

Refer to Chapter 6 of the textbook!